



Software Development Life Cycle

The SDLC is a systematic process used to plan, design, build, test, deploy, and maintain software. It ensures projects are organized, predictable, and aligned with business goals, whether following Waterfall (sequential) or Agile (iterative) models.

✨ The SDLC ensures that software projects move smoothly from **concept to reality**, balancing **business needs, technical feasibility, and user satisfaction**.



Planning

This phase defines the **project scope, objectives, resources, and timeline**. Feasibility studies (technical, financial, and operational) are conducted to ensure the project is viable.

Purpose : Establish a clear roadmap.

Deliverable : Project plan and feasibility report.



Analysis

Detailed requirements are gathered from stakeholders and documented. The goal is to understand *what* the system must achieve.

Purpose : Capture functional and non-functional requirements.

Deliverable : Software Requirement Specification (SRS).



Design

Requirements are translated into **system architecture and design blueprints**.

High-Level Design (HLD) : Architecture, modules, database.

Low-Level Design (LLD) : Workflows, APIs, logic.

Deliverable : Design Document Specification (DDS).



Development

The actual coding of the system takes place here. Developers build modules according to design documents, perform unit testing, and integrate components.

Purpose : Build the system.

Deliverable : Source code and executable application.



Testing

The system is tested for **functionality, performance, and security**.

Types : Unit, Integration, System, and User Acceptance Testing (UAT).

Purpose : Ensure quality and defect resolution.

Deliverable : Test cases, defect reports, and quality metrics.



Implementation

The system is deployed into production. Activities include installation, configuration, data migration, and user training.

Purpose : Deliver the system to end-users.

Deliverable : Live application.



Maintenance

After deployment, the system requires ongoing support. This includes bug fixes, updates, performance tuning, and adding new features.

Purpose : Keep the system reliable and up-to-date.

Deliverable : Patches, upgrades, and improved versions.



Quick Reference

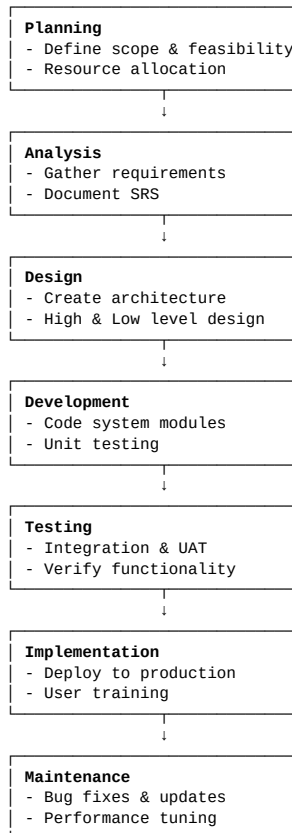
<u>Phase</u>	<u>Purpose</u>	<u>Key Deliverable</u>
Planning	Define scope, feasibility	Project Plan
Analysis	Gather requirements	SRS
Design	Create system blueprint	Design Docs
Development	Build the system	Source Code
Testing	Validate quality	Test Reports
Implementation	Deploy system	Live Application
Maintenance	Support & improve	Updates/Patches

✨ The SDLC ensures projects are **structured, predictable, and aligned with business goals**. Whether using **Waterfall (sequential)** or **Agile (iterative)** models, these seven phases remain the backbone of software development, helping teams deliver **high-quality, cost-effective, and user-centric solutions**.

Seven Phases Flow

— This flow shows the **continuous lifecycle** of software development:

Planning → Analysis → Design → Development → Testing → Implementation → Maintenance.



Each phase builds on the previous one, ensuring projects are structured, predictable, and aligned with business goals.